SHUAIXIANG ZHANG

Address: 246 Swain West, Bloomington, IN, USA
Postcode: 47405
Phone: +1 218-203-7858
Email: szh2@iu.edu

EDUCATION

Ph.D. in Experimental Particle Physics Aug. 2020 – Oct. 2026

Department of Physics, Indiana University, Bloomington, IN, USA

Thesis: *DUNE Far Detector Scintillation Photon Response Studies Using Data from the ProtoDUNE Detectors*
Advisor: Prof. Jon Urheim

B.S. in Particle Physics Sep. 2016 – Jul. 2020

Department of Modern Physics, University of Science and Technology of China, Hefei, Anhui, China

Thesis: *Study of ^{12}B Production by Stopping Cosmic Muons in the Daya Bay Experiment*
Advisors: Prof. Guangshun Huang and Dr. Jilei Xu

RESEARCH EXPERIENCES

Deep Underground Neutrino Experiment Aug. 2020 – Present

- Working on ProtoDUNE detectors and Photon Detection System (PDS)
- Advised by Prof. Jon Urheim, Department of Physics, Indiana University Bloomington
- Michel electron studies using PDS in ProtoDUNE–HD detector
- Developments of machine-learning-based fast optical propagation models for ProtoDUNE–HD, ProtoDUNE–VD, full-scale DUNE horizontal-drift far detector module
- Rayleigh scattering length measurement based on ProtoDUNE–VD
- Charge Readout Plane assembly for DUNE vertical-drift far detector module

COHERENT Experiment Jul. 2019 – Sep. 2019

- Simulation of neutrino fluxes produced by proton beam for the summer research internship
- Advised by Prof. Diana Parno, Department of Physics, Carnegie Mellon University

Undergraduate Research Dec. 2017 – Jul. 2018

- Development of large–area double–mesh micromegas detector
- Advised by Prof. Jianbei Liu, Department of Modern Physics, University of Science and Technology of China

PUBLICATIONS

- *Design and Performance of the Photon Detection System of the ProtoDUNE-HD Detector at CERN* (Working in progress)
- A. Abed Abud et al. (DUNE Collaboration) [2024]. *The DUNE Far Detector Vertical Drift Technology Technical Design Report*. JINST 19 T08004
- A. Abed Abud et al. (DUNE Collaboration) [2023]. *Identification and reconstruction of low-energy electrons in the ProtoDUNE-SP detector*. PhysRevD.107.092012
- D. Akimov et al. (COHERENT Collaboration) [2022]. *Simulating the neutrino flux from the Spallation Neutron Source for the COHERENT experiment*. PhysRevD.106.032003

TALKS & WORKSHOPS

- 2026.07.20: Presentation *"Michel Electron Optical Studies in the ProtoDUNE-HD Detector "*, DPF 2026, Fermilab, Batavia, IL, USA
- 2026.07.16: Invited seminar *"Michel Electron Studies in the DUNE Far Detector Prototypes"*, Physics Department, Brookhaven National Laboratory, Upton, NY, USA
- 2026.07.09: Invited seminar *"Michel Electron Studies in the DUNE Far Detector Prototypes"*, Department of Physics and Astronomy, Johns Hopkins University, Baltimore, MD, USA
- 2026.03.17: Presentation *"Michel Electron Studies in the DUNE Far Detector Prototype Detectors"*, APS Global Physics Summit 2026, Denver, CO, USA
- 2025.12.11: Invited seminar *"Michel Electron Studies in the DUNE Far Detector Prototype Detectors"*, Department of Physics, Duke University, Durham, NC, USA
- 2025.07.15: Presentation *"Michel Electron Studies in Prototypes of DUNE"*, New Perspective 2025, Fermilab, Batavia, IL, USA
- 2025.05.23: Plenary talk *"Michel Electron Studies Using PDS in ProtoDUNE–HD"*, DUNE collaboration meeting, Fermilab, Batavia, IL, USA
- 2025.02.24: NP04 PDS Performance Article Workshop, Valencia, Spain
- 2025.02.07: DUNE LArTPC Software & Analysis Workshop, CERN
- 2024.05.19: Invited talk (Minisymposium) *"Neural Network Based Fast Optical Simulation Method in ProtoDUNE–VD"*, DPF–PHENO 2024, Pittsburgh, PA, USA
- 2023.08.01: Poster *"Computational Graph Model for Photon Propagation in ProtoDUNE–VD"*, 14th International Neutrino Summer School, Fermilab, Batavia, IL, USA

TEACHING & OUTREACH

Introductory seminar *"Scintillators in Particle Physics Experiments"* (Indiana U)	Summer 2025
General Physics I (PHYS P201) (Indiana U): Assistant in Instruction	Fall 2023
Theoretical Mechanics (USTC): Assistant in Instruction	Fall 2019

AWARDS & HONORS

Grant-in-Aid Travel Award	Dec. 2025
Universities Research Association (URA) Visiting Scholars Award	Apr. 2025
Outstanding Winner in College Physics Experiment of Level 4 (Near–field acoustic holography)	Dec. 2018
Scholarship for Academic Excellence	2017, 2018
Silver Award of the Freshman Scholarship	2016